Zihan Fang

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Research Focus

My research develops human-centered AI to lower barriers to acquiring expertise and contributing effectively to software development. I combine large-scale repository mining, cognitive and behavioral modeling, and reinforcement learning to identify sources of difficulty, characterize how developers acquire expertise, and translate these insights into AI-supported programming systems and immersive learning environments that improve learning efficiency, programming performance, and long-term skill development.

Education

Vanderbilt University, Ph.D. in Computer Science — Aug 2023 – Present
Vanderbilt University, M.S. in Data Science; GPA: 3.9/4.0 — Aug 2021 – May 2023
Soochow University, B.S. in Information Management; GPA: 3.8/4.0 — Sep 2016 – Jun 2020

Publications

[1] Yueke Zhang, **Zihan Fang**, Yuxin Hu, Kevin Leach, Yu Huang: *Unclaimed Territory: Analyzing Subdomain Takeover Risks in GitHub Pages. In the Proceedings of the 49th IEEE/ACM International Conference on Software Engineering (**ICSE 2027**)*.

[2] **Zihan Fang**, Janice Chung, Ruijia Chen, Marcia Barnes, Yuhang Zhao, Yu Huang: *Toward Inclusive Programming Support for Novice Programmers with Dyslexia: Insights from a Comparative Study*. *In the Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (**ASE 2026**)*.

[3] Ningzhi Tang, Chaoran Chen, **Zihan Fang**, Gelei Xu, Maria Dhakal, Yiyu Shi, Collin McMillan, Yu Huang, Toby Jia-Jun Li: *Programming by Chat: A Large-Scale Behavioral Analysis of 11,579 Real-World AI-Assisted IDE Sessions*. *In the Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (**ASE 2026**)*.

[4] **Zihan Fang**, Weizhe Jiao, Robert Tairas, Yu Huang: *Novices Comprehending Functional vs. Imperative Programs: An Eye-Tracking Study*. *ACM Transactions on Software Engineering and Methodology (**TOSEM 2026**)*.

[5] Yueke Zhang, **Zihan Fang**, Kevin Leach: *Stop the Retrieval Thrash: Brain-Guided Episodic Memory for Repository-Scale Coding Agents*. *In Companion Proceedings of the ACM on Software Engineering, Vol. 3, FSE (**FSE 2026, HumanAISE**)*.

[6] Yueke Zhang, Yifan Zhang, **Zihan Fang**, Greg Trafton, Daniel Levin, Kevin Leach, Yu Huang: *ScanCoder: Leveraging Human Attention Patterns to Enhance LLMs for Code*. *In Proceedings of the ACM on Software Engineering, Vol. 3, FSE (**FSE 2026**)*.

[7] Yueke Zhang, **Zihan Fang**, Kevin Leach, Yu Huang: *Locating Software Vulnerabilities With Static Analyzers: How Far Are We? In Proceedings of the 30th International Conference on Evaluation and Assessment in Software Engineering (**EASE 2026**)*.

[8] **Zihan Fang**, Robert Wallace, Zachary Karas, Toby Jia-Jun Li, Collin McMillan, Yu Huang: *Investigating the Feasibility of Conducting Webcam-Based Eye-Tracking Studies in Code Comprehension*. *IEEE Transactions on Software Engineering (**TSE 2026**)*.

[9] **Zihan Fang**, Yueke Zhang, Thomas Zimmermann, Denae Ford, Yu Huang: *Contribution Patterns in Open Source Software for Social Good: Dynamics, Individuals, and Impact*. *In Proceedings of the ACM on Human-Computer Interaction (**CSCW 2026**)*.

[10] Yueke Zhang, **Zihan Fang**, Greg Trafton, Daniel Levin, Kevin Leach, Yu Huang: *CodeACT-R: A Cognitive Simulation Framework for Human Attention in Code Reading*. *In Companion Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2025, NIER Track**)*.

[11] **Zihan Fang**, Jiliang Li, Anda Liang, Gina R. Bai, Yu Huang: *A Comparative Study on ChatGPT and Checklist as Support Tools for Unit Testing Education*. *In Companion Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering (**FSE 2025, SEET Track**)*.

[12] **Zihan Fang**, Yu Huang: *"Math Is a Pain!": Understanding Challenges and Needs of the Machine Learning Community on Stack Overflow*. *In Proceedings of the ACM on Human-Computer Interaction. (**CSCW 2024**)*

[13] **Zihan Fang**, Madeline Endres, Thomas Zimmermann, Denae Ford, Westley Weimer, Kevin Leach, Yu Huang: *A Four-Year Study of Student Contributions to OSS vs. OSS4SG with a Lightweight Intervention*. *In Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**ESEC/FSE 2023, ACM SIGSOFT Distinguished Paper Award.**)*

Manuscripts Under Review

[1] **Zihan Fang**, Yifan Zhang, Yueke Zhang, Kevin Leach, Yu Huang: DPO-F+: *Aligning Code Repair Feedback with Developers' Preferences*. [arXiv]

[2] **Zihan Fang**, Yueke Zhang, Yu Huang: *AgentForge: An Immersive Role-Playing Platform for Learning Agentic Software*

Engineering. [arXiv]

[3] **Zihan Fang**, Yueke Zhang, Ningzhi Tang, Collin McMillan, Toby Jia-Jun Li, Yu Huang: *From Conversation to Contribution: Characterizing Coding Agent in Open-Source Software*. [arXiv]

[4] **Zihan Fang**, Ron Asoulin-Handelman, Helen Wu, Yu Huang, Gina R. Bai: *Can AI Guide Testing Without Generating Tests? A Comparative Study of AI-Assisted Unit Testing*. [arXiv]

[5] Yueke Zhang, **Zihan Fang**, Kevin Leach, Yu Huang: *GraphAlignCoder: Aligning Program and Proof Graphs for Code Generation*. [arXiv]

[6] Zachary Karas, Elham Isa, **Zihan Fang**, James Booth, Catie Chang, Yu Huang: *Thinking of it like a tree": A Naturalistic fMRI Study to Understand Recursive Coding*

[7] Yueke Zhang, **Zihan Fang**, Kevin Leach, Yu Huang: *CogAdapt: Cognition-informed Sparse Adaptation of Code LLMs*

Research and Industry Experience

Research Assistant, Vanderbilt University — Aug 2023 – Present

- Conduct research on human-centered AI for software engineering, spanning developer-feedback alignment, AI-assisted programming, and multi-agent learning environments.

Data Scientist, Tencent Inc. — Aug 2021 – Dec 2021

- Built scalable ML and data systems to support real-time identity verification and model development.

Data Analyst Intern, DiDi Inc. — Jan 2021 – Jun 2021

- Applied data analytics and machine learning to segment drivers, evaluate operational strategies, and optimize recruiting, onboarding, and business development efforts.

Teaching Experience

Teaching Assistant, Vanderbilt University — Aug 2022 – Present

- CS3892/5892 Designing Computing for Learning – Fall 2026
- DS3100 Fundamentals of Data Science – Fall 2022, Spring 2023
- CS3265 Database Management Systems – Fall 2023
- CS4278/5278 Principles of Software Engineering – Spring 2024

Awards

- ACM SIGSOFT CAPS Travel Award 2026
- Ignite Grant at Vanderbilt University 2025
- LIVE Horizon Grant at Vanderbilt University 2025
- C. F. Chen Best Paper Award in the Department of Computer Science at Vanderbilt University 2024
- Richard Bennett/Dorothy Danforth Compton Prize Scholarship at Vanderbilt University 2024
- SIGSOFT Distinguished Paper Award at FSE 2023
- NSF Student Travel Award 2023

Service

Academic Reviewing: AAAI 2026; ACL Rolling Review (ARR) 2026; Empirical Software Engineering (EMSE); SIGCSE TS 2025 (Posters, Computing Education Research).